

Note 03 · Breakage and the crushing cycle

Why small particles resist, what the breakage function describes, how CRSH and SHHD turn a setting into a product, and what Bond's law buys you in Exercise 4

1 The question this note answers

A crusher is described in MODSIM by a handful of numbers: a setting, a work index, and for the advanced models a fines fraction K and two factors α_1, α_2 . This note explains what each of them does, using the vocabulary of King chapter 5 and the Crushing lecture, so that the parameter forms of JAW1, GYRA, CRSH and SHHD stop being mysterious boxes. It ends with the one energy calculation Exercise 4 asks for.

2 A toy example before any theory

Drop a 40 mm stone into a machine that squeezes it once and let it fall out. Suppose the fragments weigh, by size: 30 % between 20 and 40 mm, 40 % between 10 and 20 mm, 20 % between 5 and 10 mm and 10 % finer than 5 mm. Written cumulatively from the fine end: 10 % finer than 5 mm, 30 % finer than 10 mm, 70 % finer than 20 mm, 100 % finer than 40 mm. That last list is a **breakage function**: the size distribution of the children of one parent. Now put the machine on a conveyor: fragments still coarser than the discharge opening are squeezed again, the rest falls through. Repeat until everything has escaped. That loop is the **crushing cycle**, and it is exactly what MODSIM's standard crusher model computes.

Concept: Why small particles are relatively stronger

Rock is brittle: it fails in tension, and cracks are where tensile stress concentrates. A large lump contains many cracks and grain boundaries; a small particle contains few. So the specific energy needed to break a particle rises steeply as it gets smaller. King's Table 5.1 gives the median fracture energy E_{50} per ore type, and the lecture writes the size dependence as

$$E_{50} = E_{\infty} \left(1 + \frac{d_0}{d_p - d_{p,\min}} \right)^{\varphi},$$

with E_{∞} the median energy of large particles (about 1 cm and above), d_0 and φ material parameters, and $d_{p,\min}$ the smallest particle that can still be fragmented (about 0.3 micrometres for quartz). Consequence: fines are expensive to make, and every millimetre removed in a crusher is cheaper than the same millimetre removed in a mill.

Concept: The breakage function $B(x; y)$

$B(x; y)$ is the cumulative mass fraction of fragments finer than x produced from a parent of size y . Three forms appear in the course.

- The two-term (Whiten or Austin) form used by CRSH and SHHD:

$$B(x; y) = (1 - K) \left(\frac{x}{y} \right)^n + K \left(\frac{x}{y} \right)^m,$$

a coarse part and a fine part; K (default 0.2) is the proportion of fines produced during a breakage event.

- Truncated distributions for shattering: the fragment curve is a size distribution with a definite top size equal to the parent; the truncated RRS is the convenient one.

- The t_{10} family for single-particle tests: t_{10} is the fraction of fragments finer than one tenth of the parent size, and from that one number the whole fragment curve ($t_2, t_4, t_{25}, \dots$) is reconstructed. t_{10} grows with the impact energy: $t_{10} = t_{10,\max} [1 - \exp(-\beta E/E_{50})]$.

In the population balance model the breakage function fills the breakage matrix B : element b_{ij} is the fraction of class j that lands in class i , and by definition all diagonal elements are 1.

Concept: The crushing cycle and the internal classification function

In each throw the crusher nips part of the material and opens again; broken particles that are now finer than the opening fall through, the rest is nipped once more. MODSIM writes, for the coarsest class, $Mm_1 = Ff_1 + c_1 Mm_1 b_{11}$: what is inside the machine equals what the feed brings plus what survived the last cycle. The share that stays for another cycle is the **internal classification function**

$$c_i = 1 - \left(\frac{d_i - k_2}{k_1 - k_2} \right)^{k_3} \quad \text{for } k_1 < d_i < k_2, \quad c_i = 0 \text{ below } k_1, \quad c_i = 1 \text{ above } k_2,$$

with $k_1 = \alpha_1 \cdot \text{CSS}$ and $k_2 = \alpha_2 \cdot \text{CSS}$. Particles finer than k_1 leave untouched; particles coarser than k_2 are always broken. CRSH defaults: $\alpha_1 = 0.653$, $\alpha_2 = 1.6$. SHHD (short-head, tertiary duty) uses $\alpha_2 = 2$. With a CSS of 25 mm, CRSH lets everything under 16 mm pass and breaks everything over 40 mm; SHHD breaks everything over 50 mm.

3 The crusher models of MODSIM, sorted by what drives them

Model	What sets the product	Parameters on the form
JAW1, JAW2	standard curve scaled by the OSS; feed size does not enter	OSS, impact work index
GYRA	standard curve scaled by the OSS, adjusted for ore type	OSS, ore type, work index
EMJC	power law from the gap up to $r_{\max}$	gap, $r_{\max}$, exponent m , work index
CRS1	standard Symons curve from the CSS (preliminary only)	CSS
CRSH	crushing cycle: classification plus two-term breakage	CSS (or OSS for jaw/gyratory d
SHHD	same cycle with Karra's short-head parameters	CSS, K , W_i , α_1 , α_2

Two facts to carry into Exercise 3. First, the primary models produce a product that does not depend on the feed size distribution at all (only capped at the largest feed size), and about 83 % of it is finer than the OSS with nothing coarser than 1.25 times the OSS. Second, none of these models has a capacity term: raise the feed by 50 % and the product curves stay where they were, while the power estimate scales with tonnage. The capacity check is yours, against King's Tables 5.4 to 5.7. The lecture's closing advice: when unsure, take the simplest model.

Concept: Energy: three laws, one Bond formula

Kick, Bond and Rittinger are the same differential energy-size relation with different exponents; in practice crushing sits in the Kick range, ordinary grinding in the Bond range and fine grinding in the Rittinger range. MODSIM reports crusher energy with a Bond-type relation in which a crusher index replaces the work index, which is why every crusher form

asks for an impact work index. Bond's law, with sizes in *micrometres*:

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \quad [\text{kWh/t}].$$

4 The numbers of Exercise 4

Work index 12 kWh/t. The primary mill must take the crusher-stage product (d_{80} about 5 mm) to $d_{80} = 0.1$ mm:

$$W_{\text{mill}} = 10 \times 12 \times \left(\frac{1}{\sqrt{100}} - \frac{1}{\sqrt{5000}} \right) = 120 \times (0.1000 - 0.0141) = 10.3 \text{ kWh/t}.$$

The crusher stage, from a crusher feed d_{80} of about 25 mm to about 7 mm, on the tonnes it actually crushes:

$$W_{\text{crusher}} = 120 \times \left(\frac{1}{\sqrt{7000}} - \frac{1}{\sqrt{25500}} \right) = 120 \times (0.01195 - 0.00626) = 0.68 \text{ kWh per crushed tonne},$$

which is about $0.68 \times 315/600 = 0.36$ kWh per tonne of circuit feed in the open circuit. The mill costs twenty to thirty times the crusher. That ratio is the engineering content of the sheet's sentence "from an energy point of view it is often best to crush as far as possible".

In one sentence

Small particles resist breakage, a breakage function describes the children of one parent, a crusher repeats select-and-break until the fragments escape through the setting, and Bond's law with sizes in micrometres turns the resulting d_{80} values into the kilowatt-hours the seminar will ask about.

Watch out / typical mistake

- Bond with millimetres instead of micrometres inflates the energy about thirtyfold. Convert first: 5 mm is 5000 micrometres.
- CSS typed into a primary model. JAW1, JAW2 and GYRA want the open-side setting; CRSH reads its single setting as CSS for cones and OSS for jaws and gyratories.
- Concluding "the crusher can take it" from a simulation. No MODSIM crusher model checks capacity; only the tables do. In closed circuit the tonnage to check includes the recirculating load.
- Metres in the GUI: a 25 mm CSS is 0.025.

Try it yourself (keyboard in hand)

1. Open **BreakageModels.xls** (module 2 material) and move t_{10} from 10 to 30 %. Watch the whole fragment curve lift. Then open **SecCrushProd.xls** and change the CSS: the k_1 and k_2 limits move with it.
2. In MODSIM, put a CRSH and a SHHD side by side on the same 100 t/h RRS feed of Exercise 3, both at CSS 25 mm, defaults otherwise. Compare the products: SHHD breaks up to 50 mm before letting go, CRSH up to 40 mm.
3. Set K to 0.1 and then to 0.4 in CRSH and rerun. The fine end of the product moves; the coarse end barely does. That is what "proportion of fines produced during a breakage event" means.

4. Repeat the Bond calculation of Exercise 4 with a mill feed d_{80} of 8 mm instead of 5 mm. The mill term rises from 10.3 to about 10.7 kWh/t: crushing two more millimetres saved that difference.

Glossary: in plain words

OSS, CSS open- and closed-side setting: the widest and the narrowest gap of the crushing chamber during a throw.

Breakage function the size distribution of the fragments of one parent particle.

Selection, classification which particles get broken in a cycle; in a crusher this is the internal classification function.

t_{10} fraction of fragments finer than a tenth of the parent size; the anchor of the single-particle breakage description.

Work index W_i Bond's material constant in kWh/t; 12 for the Kiruna ore in Exercise 4.

Choked crushing feeding a crusher faster than it can discharge at its setting; the sheets forbid any risk of it.